

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

张杰博士生于 1990 年 10 月 20 日，现任北京大学计算机学院助理教授、特聘研究员、博士生导师，获得优秀青年科学基金海外项目（海外优青）、北京市自然科学基金青年重点项目（原北京杰青）、华为奥林帕斯先锋奖、英特尔学术英才计划荣誉学者、ACM SIGCSE 新星奖、博雅青年学者。他长期从事存储系统和专用处理器的研究和设计，致力于从计算机体系结构层面出发，解决大数据和人工智能时代对于高性能存储系统的需求，突破冯诺依曼体系结构下数据迁移的瓶颈以及内存墙的限制。负责和参与科技部重点研发项目（课题负责人）、国家自然科学基金重点项目（北大负责人）、国家科技重大专项和面上项目，得到华为、江波龙、三星电子、海力士、英特尔和西部数据的资助。自 2016 年起，在计算机国际会议及期刊上发表了 80 余篇论文，包括 ISCA、OSDI、SOSP、HPCA、MICRO、ASPLOS、FAST、ATC、NSDI、DAC 和 Eurosys。2025 年入选 HPCA 名人堂（首位 90 后成员）。在计算机系统结构领域 90 后学者中，发表顶级会议论文数排名全球第一（CSRankings, 2016-2026）。亚洲历史上第三位在所有体系结构和系统顶级会议上发表论文的学者。论文获 2019 年韩国计算机大会（KCC）最佳演讲论文奖和 2019 年非易失性内存研讨会（NVMW）优秀论文提名奖。在存储领域的研究成果被列为 KAIST50 年重大突破之一，并被四十余家新闻媒体报道。

工作经历

北京大学，北京，中国

计算机学院

KAIST, Daejeon, Korea

Electrical Engineering (Computing Division)

助理教授，博导

2021 年 7 月–至今

博士后

2020 年 3 月 –2021 年 5 月

教育经历

Yonsei University, Incheon, Korea

PhD in Engineering

University of Texas at Dallas, Richardson, Texas

PhD in Computer Engineering (transfer to Korea)

University of Texas at Dallas, Richardson, Texas

Master of Science in Electrical Engineering

Nanjing University of Posts and Telecommunications, Nanjing, China

BS in Communication Engineering (computer communication)

Advisor: Dr. Myoungsoo Jung

August 2015 – Feb 2020

Advisor: Dr. Myoungsoo Jung

August 2014 – August 2015

Advisor: Dr. Myoungsoo Jung

August 2012 – May 2014

September 2008 – July 2012

论文发表情况

2027

Eurosys

Beyond Overlap: Hierarchical Graph Abstraction for Out-of-Core Vector Search

CCF-A 会议

Lizheng Chen, Pinhuan Huang, Shaonan Ma, [Jie Zhang](#), Qing Wang, Zhengyi Yang, Heng Zhang, Mingxing Zhang, Fan Yang, Guihai Chen, Chen Tian, Chengying Huan

22th ACM European Conference on Computer Systems

2026

TACO

Cambricon-FlexLLM: A Flexible Chiplet-Based Hybrid Architecture for On-Device 70B LLM Inference

CCF-A 期刊

Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, [Jie Zhang](#), Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi Chen
Transactions on Architecture and Code Optimizations

MICRO

LUTHE: Breaking the Memory Wall of High-Precision Encrypted LLM Nonlinearities via

Last update: August 17th, 2026

GPU-CSD Co-Design**CCF-A 会议***Xiurui Pan, Guoci Chen, Guangyu Sun, Mingzhe Zhang, [Jie Zhang](#)***The 58th Annual IEEE/ACM International Symposium on Microarchitecture***SC****TPage: Transparent Page Management for CXL-Based Tiered Memory Systems****CCF-A 会议***Maoxin Ye, Suzhen Wu, [Jie Zhang](#), Bo Mao**International Conference for High Performance Computing, Networking, Storage and Analysis***ICS****Latency-SLO-Aware Memory Offloading for Large Language Model Inference****CCF-B 会议***Chenxiang Ma, Hanyu Zhao, Zhisheng Ye, Zehua Yang, Tianhao Fu, Jiaxun Han, [Jie Zhang](#), Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Yong Li, Diyu Zhou**ACM International Conference on Supercomputing***HPDC****BCCE: Block-Centric GPU Co-Design for Real-Time Range-Top-K Query at Scale****CCF-A 会议***Chengying Huan, Ziheng Meng, Zhengyi Yang, Yongchao Liu, [Jie Zhang](#), Qing Wang, Jing Wang, Shaonan Ma, Zhibin Wang, Mingxing Zhang, Rong Gu, Baokun Wang, Guihai Chen, Chen Tian**ACM International Symposium on High-Performance Parallel and Distributed Computing***OSDI****Espresso: Constructing Cost-Efficient CXL JBOF via Inter-SSD Computing Resource****CCF-A 会议****Sharing***Shushu Yi, Yuda An, Li Peng, Xiurui Pan, Qiao Li, Jieming Yin, Guangyan Zhang, Wenfei Wu, Chenxi Wang, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Ke Zhou, [Jie Zhang](#)***USENIX Symposium on Operating Systems Design and Implementation***OSDI****Merlin: An Efficient Adaptive Cache Eviction Algorithm via Fine-Grained Characterization****CCF-A 会议***Liuji Li, Jinhao Guo, Yi Fan, Jianyu Wu, Zhenling Wang, [Jie Zhang](#), Yuval Tamir, Xiaolin Wang, Yingwei Luo, Diyu Zhou**USENIX Symposium on Operating Systems Design and Implementation***OSDI****MultiLane: Eliminating Centralized Bottlenecks in User-space Network Stack****CCF-A 会议***Kang Hu, Shuqi Dong, Chuandong Li, Ran Yi, Zonghao Zhang, [Jie Zhang](#), Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou**USENIX Symposium on Operating Systems Design and Implementation***ICLR****LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences****CCF-A 会议***Wenbo Wu, Qingyi Si, Xiurui Pan, Ye Wang, [Jie Zhang](#)***The Fourteenth International Conference on Learning Representations***DAC****WILL: Write Invalidation Skipping for QLC NAND by Leveraging Data Lifespan and Latency Awareness****CCF-A 会议***Beichen Ning, Yina Lv, Tianyu Ren, Bin Gao, Xiangyu Yao, [Jie Zhang](#), Qiao Li and Chun Jason Xue**Design Automation Conference***DAC****Cachence: Fine-Grained Cache Partitioning in Both Time and Space**

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

CCF-A 会议	<i>Liujia Li, Yuanlong Li, Yiming Yao, Jianyu Wu, Yi Fan, Jinhao Guo, Liren Zhu, Jie Zhang, Xiaolin Wang, Yingwei Luo, Zhenlin Wang, Diyu Zhou</i> <i>Design Automation Conference</i>
FAST	Xerxes: Extensive Exploration of Scalable Hardware Systems with CXL-Based Simulation Framework
CCF-A 会议	<i>Yuda An, Shushu Yi, Bo Mao, Qiao Li, Mingzhe Zhang, Diyu Zhou, Ke Zhou, Nong Xiao, Guangyu Sun, Yingwei Luo, Jie Zhang*</i> <i>The USENIX Conference on File and Storage Technologies</i>
HPCA	AutoGNN: End-to-End Hardware-Driven Graph Preprocessing for Enhanced GNN Performance
CCF-A 会议	<i>Seungkwan Kang, Seungjun Lee, Donghyun Gouk, Miryeong Kwon, Hyunkyu Choi, Junhyeok Jang, Sangwon Lee, Huiwon Choi, Jie Zhang, Wonil Choi, Mahmut T Kandemir, Myoungsoo Jung</i> <i>IEEE International Symposium on High-Performance Computer Architecture</i>
Eurosys	ColdCode: Cold Data Encoding for Enhanced Reliability and Lifetime in 3D NAND Flash
CCF-A 会议	<i>Qiao Li, Shangyu Wu, Zheng Wan, Yufei Cui, Jie Zhang*, Chun Jason Xue</i> <i>21th ACM European Conference on Computer Systems</i>
2025	
TCAD	CXL-DMSim: A Full-System CXL Disaggregated Memory Simulator With Comprehensive Silicon Validation
CCF-A 期刊	<i>Yanjing Wang, Lizhou Wu, Wentao Hong, Yang Ou, Zicong Wang, Sunfeng Gao, Jie Zhang, Sheng Ma, Dezun Dong, Xingyun Qi, Mingche Lai, Nong Xiao</i> <i>ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems</i>
PACT	Exploring Memory Tiering Systems in the CXL Era via FPGA-based Emulation and Device-Side Management
CCF-B 会议	<i>Yiqi Chen, Xiping Dong, Zhe Zhou, Zhao Wang, Jie Zhang, Guangyu Sun,</i> <i>International Conference on Parallel Architecture and Compilation Techniques</i>
NEWTON	Ultrafast and Reliable Domain-Wall and Skyrmion Logic in a Chirally Coupled Ferrimagnet
CELL 子刊	<i>Yifei Ma, Dihua Wu, Fengbo Yan, Xiaoxiao Fang, Peixin Qin, Leran Wang, Li Liu, Laichuan Shen, Zhiqi Liu, Wenyun Yang, Jie Zhang, Yanglong Hou, Yan Zhou, Feng Luo, Jinbo Yang, Kaiming Cai, Shuai Ning, Zhaochu Luo</i> <i>Cell Press NEWTON</i>
SOSP	Aeolia: Fast and Secure Userspace Interrupt-Based Storage Stack
CCF-A 会议	<i>Chuangdong Li, Ran Yi, Zonghao Zhang, Jing Liu, Changwoo Min, Jie Zhang, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou</i> <i>ACM Symposium on Operating Systems Principles</i>
HotStorage	SPDK+: Low Latency or High Power Efficiency? We Take Both <i>Endian Li, Shushu Yi, Li Peng, Qiao Li, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Bo Mao,</i>

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

Yingwei Luo, Ke Zhou, **Jie Zhang***

The 17th ACM Workshop on Hot Topics in Storage and File Systems

ICDE

BL-Tree: The Best of Both Worlds by combining B+-Tree on Top and LSM-Tree on Bottom

CCF-A 会议

Maoxin Ye, Zuocheng Wang, Shengzhe Wang, Jiahong Chen, Chunfeng Du, **Jie Zhang**, Ke Zhou, Suzhen Wu, Bo Mao

IEEE International Conference on Data Engineering

ToS

Enhancing the Performance of Next-Generation SSD Arrays: A Holistic Approach

CCF-A 期刊

Jie Zhang*, Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung

ACM Transactions on Storage

TCAD

FlexHMB: A Flexible HMB Design towards Bufferless Mobile Flash

CCF-A 期刊

Yong Peng, Li Peng, Shaocong Sun, Lujia Yin, Qiao Li, **Jie Zhang***

ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems

ISCA

XHarvest: Rethinking High-Performance and Cost-Efficient SSD Architecture with

CCF-A 会议

CXL-Driven Harvesting

Li Peng, Wenbo Wu, Shushu Yi, Xianzhang Chen, Chenxi Wang, Shengwen Liang, Zhe Wang, Nong Xiao, Qiao Li, Mingzhe Zhang, **Jie Zhang***

IEEE/ACM International Symposium on Computer Architecture

ISCA

ArtMem: Adaptive Migration in Reinforcement Learning-Enabled Tiered Memory

CCF-A 会议

Xinyue Yi, Hongchao Du, Yu Wang, **Jie Zhang**, Qiao Li, Chun Jason Xue

IEEE/ACM International Symposium on Computer Architecture

Eurosys

Daredevil: Rescue Your Flash Storage from Inflexible Kernel Storage Stack

CCF-A 会议

Junzhe Li, Ran Shu, Jiayi Lin, Qingyu Zhang, Ziyue Yang, **Jie Zhang**, Yongqiang Xiong, Chenxiong Qian

20th ACM European Conference on Computer Systems

NSDI

Beehive: A Scalable Disaggregated Memory Runtime Exploiting Asynchrony of

CCF-A 会议

Multithreaded Programs

Quanxi Li, Hong Huang, Ying Liu, Yanwen Xia, **Jie Zhang**, Mosong Zhou, Xiaobing Feng, Huimin Cui, Quan Chen, Yizhou Shan, Chenxi Wang

22nd USENIX Symposium on Networked Systems Design and Implementation

HPCA

InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM

CCF-A 会议

Inference

Xiurui Pan, Endian Li, Qiao Li, Shengwen Liang, Yizhou Shan, Ke Zhou, Yingwei Luo, Xiaolin Wang, **Jie Zhang***

IEEE International Symposium on High-Performance Computer Architecture

HPCA

NeuVSA: A Unified and Efficient Accelerator for Neural Vector Search

CCF-A 会议

Ziming Yuan, Lei Dai, Wen Li, **Jie Zhang**, Shengwen Liang, Ying Wang, Cheng Liu, Huawei Li, Jiafeng Guo, Peng Wang, Renhai Chen, Gong Zhang

HPCA

Criticality-Aware Instruction-Centric Bandwidth Partitioning for Data Center Applications

CCF-A 会议

Liren Zhu, Liujia Li, Jianyu Wu, Yiming Yao, Zhan Shi, **Jie Zhang**, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Diyu Zhou

IEEE International Symposium on High-Performance Computer Architecture

2024

SOSP

BIZA: Design of Self-Governing Block-Interface ZNS AFA for Endurance and Performance

CCF-A 会议

Shushu Yi, Shaocong Sun, Li Peng, Yingbo Sun, Ming-Chang Yang, Zhichao Cao, Qiao Li, Myoungsoo Jung, Ke Zhou, **Jie Zhang***

ACM Symposium on Operating Systems Principles

MICRO

NeoMem: Hardware/Software Co-Design for CXL-Native Memory Tiering

CCF-A 会议

Zhe Zhou, Yiqi Chen, Tao Zhang, Yang Wang, Ran Shu, Shuotao Xu, Peng Cheng, Lei Qu, Yongqiang Xiong, **Jie Zhang**, Guangyu SunThe 57th Annual IEEE/ACM International Symposium on Microarchitecture

MICRO

Cambricon-Ilm: A chiplet-based hybrid architecture for on-device Inference of 70B LLM

CCF-A 会议

Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, **Jie Zhang**, Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi ChenThe 57th Annual IEEE/ACM International Symposium on Microarchitecture

ToS

Extremely-Compressed SSDs with I/O Behavior Prediction

CCF-A 期刊

Xiangyu Yao, Qiao Li, Kaihuan Lin, Xinbiao Gan, **Jie Zhang**, Congming Gao, Zhirong Shen, Quanqing Xu, Chuanhui Yang, Jason Xue

ACM Transactions on Storage

TC

Land of Oz: Resolving Orderless Writes in Zoned Namespace SSDs

CCF-A 期刊

Yingjia Wang, You Zhou, Fei Wu, **Jie Zhang**, Ming-Chang Yang

IEEE Transactions on Computers

USENIX ATC

ScalaAFA: Constructing User-Space All-Flash Array Engine with Holistic Designs

CCF-A 会议

Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung, **Jie Zhang***
USENIX Annual Technical Conference

USENIX ATC

ScalaCache: Scalable User-Space Page Cache Management with Software-Hardware Coordination

CCF-A 会议

Li Peng, Yuda An, You Zhou, Chenxi Wang, Qiao Li, Chuanning Chen, **Jie Zhang***

USENIX Annual Technical Conference

Arxiv

From RDMA to RDCA: Toward High-Speed Last Mile of Data Center Networks Using Remote Direct Cache AccessQiang Li, Qiao Xiang, Derui Liu, Yuxin Wang, Haonan Qiu, Xiaoliang Wang, **Jie Zhang**, Ridi Wen, Haohao Song, Gexiao Tian, Chenyang Huang, Lulu Chen, Shaozong Liu, Yaohui Wu, Zhiwu Wu, Zicheng Luo, Yuchao Shao, Chao Han, Zongjie WU, Jianbo Dong, Zheng Cao, Jinbo WU, Jiwu Shu, Jiesheng Wu

Arxiv

TACO

Characterizing and Optimizing LDPC Performance on 3D NAND Flash Memories

CCF-A 期刊

Qiao Li, Yu Chen, Guanyu Wu, Yajuan Du, Min Ye, Xinbiao Gan, [Jie Zhang](#), Zhirong Shen, Jiwu Shu, Chun Jason Xue

ACM Transactions on Architecture and Code Optimization

ISCA

Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning

CCF-A 会议

AggregationXiurui Pan, Yuda An, Shengwen Liang, Bo Mao, Mingzhe Zhang, Qiao Li, Myoungsoo Jung, [Jie Zhang*](#)

IEEE/ACM International Symposium on Computer Architecture

ASPLOS

Achieving Near-Zero Read Retry for 3D NAND Flash Memory

CCF-A 会议

Min Ye, Qiao Li, Yina Lv, [Jie Zhang](#), Tianyu Ren, Daniel Wen, Tei-Wei Kuo, Chun Jason Xue

ACM International Conference on Architectural Support for Programming Languages and Operating Systems

HPCA

StreamPIM: Streaming Matrix Computation in Racetrack Memory

CCF-A 会议

Yuda An, Yunxiao Tang, Shushu Yi, Li Peng, Xiurui Pan, Guangyu Sun, Zhaochu Luo, Qiao Li, [Jie Zhang*](#)

IEEE International Symposium on High-Performance Computer Architecture

HPCA

BeaconGNN: Large-Scale GNN Acceleration with Asynchronous In-Storage Computing

CCF-A 会议

Yuyue Wang, Xiurui Pan, Yuda An, [Jie Zhang*](#), Glenn Reinman*

IEEE International Symposium on High-Performance Computer Architecture

HPCA

LearnedFTL: A Learning-based Page-level FTL for Reducing Double Reads in Flash-based SSDs

CCF-A 会议

Shengzhe Wang, Zihang Lin, Suzhen Wu, Hong Jiang, [Jie Zhang](#), Bo Mao

IEEE International Symposium on High-Performance Computer Architecture

HPCA

Midas Touch: Invalid-Data Assisted Reliability and Performance Boost for 3D High-Density Flash

CCF-A 会议

QiaoLi, Hongyang Dang, Zheng Wan, Congming Gao, Min Ye, [Jie Zhang](#), Tei-Wei Kuo, Chun Jason Xue

IEEE International Symposium on High-Performance Computer Architecture

2023

NVMW

Optimizations of Linux Software RAID System for Next-Generation StorageShushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, [Jie Zhang*](#)The 14th Annual Non-volatile Memories Workshop

SAC

BcBench: Exploring Throughput Processor Designs based on Blockchain BenchmarkingXiurui Pan, Yue Chen, Shushu Yi, [Jie Zhang*](#)

The 38 th ACM/SIGAPP Symposium on Applied Computing	
CAL	Intelligent SSD Firmware for Zero-Overhead Journaling
SCI 3 	Hanyeoreum Bae, Donghyun Gouk, Seungjun Lee, Jiseon Kim, Sungjoon Koh, Jie Zhang* , Myoungsoo Jung*, IEEE Computer Architecture Letters (CAL)
2022	
HotStorage	ScalaRAID: Optimizing Linux Software RAID System for Next-Generation Storage Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, Jie Zhang* 14 th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 22)
THPC	Survey on Storage-Accelerator Data Movement Zixuan Zhou, Shushu Yi, Jie Zhang* CCF Transaction on High Performance Computing
NVMW	Integrating New Photonic-Based Heterogeneous Memory into Throughput Accelerators Jie Zhang , Myoungsoo Jung, The 13 th Annual Non-volatile Memories Workshop
NVMW	HAMS: Hardware Automated Memory-over-Storage for Large-scale Memory Expansion Jie Zhang , Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim, Mahmut Taylan Kandemir, Myoungsoo Jung The 13 th Annual Non-volatile Memories Workshop
2021	
MICRO CCF-A 	Ohm-GPU: Integrating New Optical Network and Heterogeneous Memory into GPU Multi-Processors Jie Zhang , Myoungsoo Jung, The 54 th Annual IEEE/ACM International Symposium on Microarchitecture
ISCA CCF-A 	Revamping Storage Class Memory with Hardware Automated Memory-Over-Storage Solution Jie Zhang , Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung, The IEEE/ACM International Symposium on Computer Architecture
NVMW	Architecting Throughput Processors with New Flash Jie Zhang , Myoungsoo Jung, The 12 th Annual Non-volatile Memories Workshop
NVMW	DRAM-less Accelerator for Energy Efficient Data Processing Jie Zhang , Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung The 12 th Annual Non-volatile Memories Workshop

NVMW **Manycore-Based Scalable SSD Architecture Towards One and More Million IOPS**
Jie Zhang, Miryeong Kwon, Michael Swift, Myoungsoo Jung,
The 12th Annual Non-volatile Memories Workshop

2020

ISCA **ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis**
CCF-A 会议 *Jie Zhang, Myoungsoo Jung,*
The IEEE/ACM International Symposium on Computer Architecture

FAST **Scalable Parallel Flash Firmware for Many-core Architectures**
CCF-A 会议 *Jie Zhang, Miryeong Kwon, Michael Swift, Myoungsoo Jung,*
The 18th USENIX Conference on File and Storage Technologies

HPCA **DRAM-less: Hardware Acceleration of Data Processing with New Memory**
CCF-A 会议 *Jie Zhang, Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung*
26th IEEE International Symposium on High-Performance Computer Architecture

ISPASS **Data Direct I/O Characterization for Future I/O System Exploration**
CCF-C 会议 *Mohammad Alian, Yifan Yuan, Jie Zhang, Ren Wang, Myoungsoo Jung, Nam Sung Kim*
The IEEE International Symposium on Performance Analysis of Systems and Software

CAL **FastDrain: Removing Page Victimization Overheads in NVMe Storage Stack**
SCI-3 区 *Jie Zhang, Miryeong Kwon, Sanghyun Han, Nam Sung Kim, Mahmut Kandemir and*
Myoungsoo Jung
IEEE Computer Architecture Letters (CAL)

2019

HPCA **FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads**
CCF-A 会议 *Jie Zhang, Myoungsoo Jung, Mahmut Kandemir,*
25th IEEE International Symposium on High-Performance Computer Architecture

IISWC **Faster than Flash: An In-Depth Study of System Challenges for Emerging Ultra-Low Latency SSDs**
Sungjoon Koh, Junkyeok Jang, Changrim Lee, Miryeong Kwon, Jie Zhang, Myoungsoo Jung,
The 2019 IEEE International Symposium on Workload Characterization

DAC **FlashGPU: Placing New Flash Next to GPU Cores**
CCF-A 会议 *Jie Zhang, Miryeong Kwon, Hyojong Kim, Hyesoon Kim, Myoungsoo Jung,*
The 56th Design Automation Conference (DAC)

TPDS **Exploring Fault-Tolerant Erasure Codes for Scalable All-Flash Array Clusters**
CCF-A 期刊 *Sungjoon Koh, Jie Zhang, Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim,*
Myoungsoo Jung,

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

IEEE Transactions on Parallel and Distributed Systems (TPDS)

NVMW

Addressing Fast-Detrapping for Reliable 3D NAND Flash Design

Mustafa Shihab, [Jie Zhang](#), Myoungsoo Jung, Mahmut Kandemir,

10th Annual Non-Volatile Memories Workshop -- Nominated as Memorable Paper Award

KCC

Maximizing GPU Cache Utilization with Adjustable Cache Line Management

[Jie Zhang](#), Myoungsoo Jung,

Korean Computer Congress (KCC), 2019 -- Nominated as Excellent Paper Award

2018

OSDI

FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for

CCF-A 会议

Ultra-Low Latency SSDs

[Jie Zhang](#), Miryeong Kwon, Donghyun Gouk, Changlim Lee, Mohammad Alian, Myoungjun

Chun, Mahmut Kandemir, Nam Sung Kim, Jihong Kim, Myoungsoo Jung,

13th USENIX Symposium on Operating Systems Design and Implementation

MICRO

Amber: Enabling Precise Full-System Simulation with Detailed Modeling of All SSD

CCF-A 会议

Resources

Donghyun Gouk, Miryeong Kwon, [Jie Zhang](#), Sungjoon Koh, Wonil Choi, Nam Sung Kim,

Mahmut Kandemir, Myoungsoo Jung,

The 51st Annual IEEE/ACM International Symposium on Microarchitecture

TACO

ReveNAND: A Fast-Drift Aware Resilient 3D NAND Flash Design

CCF-B 期刊

Mustafa Shihab, [Jie Zhang](#), Myoungsoo Jung, Mahmut Kandemir,

ACM Transactions on Architecture and Code Optimization (TACO), 2018

Eurosys

FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems

CCF-A 会议

[Jie Zhang](#), Myoungsoo Jung,

The European Conference on Computer Systems (EuroSys), 2018

IPDPS

CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for

CCF-B 会议

GPUs

[Jie Zhang](#), Shuwen Gao, Nam Sung Kim, Myoungsoo Jung,

32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2018

2017

CAL

SimpleSSD: Modeling Solid State Drive for Holistic System Simulation

SCI-3 区

Myoungsoo Jung, [Jie Zhang](#), Ahmed Abulila, Miryeong Kwon, Narges Shahidi, John Shalf,

Nam Sung Kim and Mahmut Kandemir,

IEEE Computer Architecture Letters (CAL), 2017

IISWC

Understanding System Characteristics of Online Erasure Coding on Scalable, Distributed

and Large-Scale SSD Array Systems

Sungjoon Koh, [Jie Zhang](#), Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,

IEEE International Symposium on Workload Characterization (IISWC), 2017

IISWC

TraceTracker: Hardware/Software Co-Evaluation for Large-Scale I/O Workload Reconstruction

Miryeong Kwon, [Jie Zhang](#), Gyuyoung Park, Wonil Choi, David Donofrio, John Shalf, Mahmut Kandemir, Myoungsoo Jung,

IEEE International Symposium on Workload Characterization (IISWC), 2017

NPC

An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing

[Jie Zhang](#), Myoungsoo Jung,

IFIP International Conference on Network and Parallel Computing (NPC), 2017

NPC/IJPP

Enabling Realistic Logical Device Interface and Driver for NVM Express Enabled Full System Simulations

Donghyun Gouk, [Jie Zhang](#), Myoungsoo Jung,

IFIP International Conference on Network and Parallel Computing (NPC) and Invited for International Journal of Parallel Programming (IJPP), 2017

CCF-C 会议

CCF-C 会议

2016

HPCA

DUANG: Fast and Lightweight Page Migration in Asymmetric Memory Systems

CCF-A 会议

Hao Wang, [Jie Zhang](#), Gieseok Park, Sharmila Shridhar, Myoungsoo Jung, Nam Sung Kim, *IEEE Symposium on High Performance Computer Architecture (HPCA), 2016*

ASBD

A Study for Block-level I/O Trace Reconstruction on All-Flash Arrays

Miryeong Kwon, [Jie Zhang](#), Gyuyoung Park, Myoungsoo Jung,

Workshop on Architectures and Systems for Big Data (ASBD@ISCA), 2016

NVMSA

An In-Depth Study of Next Generation Interface for Emerging Non-Volatile Memories

Wonil Choi, [Jie Zhang](#), Shuwen Gao, Jaesoo Lee, Myoungsoo Jung, Mahmut Kandemir, *IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA), 2016*

INFLOW

ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture

[Jie Zhang](#), Miryeong Kwon, Chanyoung Park, Myoungsoo Jung, Songkuk Kim,

USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

INFLOW

Couture: Tailoring STT-MRAM for Persistent Main Memory

Mustafa Shihab, [Jie Zhang](#), Shuwen Gao, Josep Sloan, Myoungsoo Jung,

USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

2015

- ASBD** **CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing**
Jie Zhang, Damian Szmulewicz, Erick Macias, Myoungsoo Jung,
The 5th Workshop on Architecture and System for Big Data (ASBD), 2015
- PACT** **NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing**
CCF-B 会议 *Jie Zhang, David Donofrio, John Shalf, Myoungsoo Jung,*
The 24th International Conference on Parallel Architecture and Compilation Techniques
- ICCD** **OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory**
CCF-B 会议 **Characterization**
Jie Zhang, Giese Park, David Donofrio, Mustafa Shihab, John Shalf and Myoungsoo Jung,
The 33rd International Conference on Computer Design (ICCD), 2015
- PACT-SRC** **Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing**
Jie Zhang, David Donofrio, John Shalf and Myoungsoo Jung,
International Conference on parallel Architecture and Compilation Techniques (PACT) –
ACM SRC 2nd Runner Award, 2015
- FAST-WiP** **Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing**
Jie Zhang and Myoungsoo Jung,
USENIX Conference on File and Storage Technologies Working in Progress (FAST WiP), 2015

2014

- HotStorage** **Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration**
Jie Zhang, Myoungsoo Jung,
6th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 14), 2014

专利发表情况

- “GPU 原生的高性能全闪存阵列软硬件设计”，易舒舒，张杰 (pending)
- “一种面向大模型推理的旁路层转换层的计算型存储设备及数据处理方法”，张杰，潘修睿，周可 (pending)
- “一种基于主机端资源借用的成本效益型固态硬盘及主机”，张杰，彭力，武文博，易舒舒，陈咸彰，王晨曦，梁胜文，王喆，李乔，张明喆 (pending)
- “基于硬件一致性互联实现盘间资源共享的全闪存阵列”，易舒舒，张杰，安昱达，彭力，李乔，尹捷明，吴文斐，周迪宇，王振林，汪小林，罗英伟，周可 (pending)
- “基于软硬协同设计的全闪存阵列”，张杰，易舒舒，潘修睿，徐一茗，李乔，李强，王晨曦，毛波，Myoungsoo Jung (pending)
- “Memory controlling device and computing device including the same”，Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, Jie Zhang, America (US20190171566A1)
- “Flash-based accelerator and computing device including the same”，Myoungsoo Jung, Jie Zhang, America

张杰

tel: +8619910031394 | email: jiez@pku.edu.cn

(US10824341B2, US20180321859, US20170285968)

- “基于闪存的加速器和包含其的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN107291424)
- “基于闪存的加速器及包括该加速器的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN109460369)
- “Resistance switching memory-based accelerator”, Myoungsoo Jung, Gyuyoung PARK, Jie Zhang, America (US20180321880A1)
- “PARALLEL PROCESSING UNIT, COMPUTING DEVICE INCLUDING THE SAME, AND THREAD SCHEDULING METHOD”, Jie Zhang, Myoungsoo Jung, America (WO2018021620)
- “MEMORY CONTROL APPARATUS AND COMPUTING DEVICE INCLUDING SAME”, JUNG MYOUNGSOO, GOUK DONGHYUN, KWON MIRYEONG, KOH SUNGJOON, 정명수, JIE ZHANG, 국동현, 권미령, 고성준 장지에, Korea (KR1020180126267)
- “COMPUTING DEVICE, METHOD OF PROCESSING INPUT/OUTPUT REQUEST, AND RECORDING MEDIUM”, Jie Zhang, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, America (pending)
- “FLASH-BASED COPROCESSOR”, Jie Zhang, Myoungsoo Jung, America (pending)
- “FLASH STORAGE DEVICE AND METHOD OF SCHEDULING PAGE VICTIMIZATION”, Jie Zhang, Myoungsoo Jung, America (pending)

学术活动情况

Journal Guest Editor

- IEEE MICRO Special Issue

Journal Paper Reviewer

- IEEE Transactions on Computer
- ACM Transactions on Storage
- ACM Transactions on Architecture and Code Optimization
- ACM Transactions on Computer Systems
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Computer Architecture Letters
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Conference Program Committee Member

- Eurosys'27
- APSys'26
- OSDI'26
- MICRO'25'24
- ISCA'26'25'24
- HPCA'27'26'25'24
- ATC'25'23
- ISPASS'26'25
- USENIX ATC'23
- ChinaSys'24'22
- SAC'22

演讲情况

- Invited talk, “Storage Architecture Design in AI Era”, Nanjing University of Posts and Telecommunications, 2026
- Invited talk, “Progress in Near-Data Processing for AI”, CCF SYS, 2025
- Invited talk, “Storage Architecture Design in AI Era”, CCF China Storage Conference, 2025

Last update: August 17th, 2026

- Invited talk, “Memory System Design based on Coherence Interconnects”, HPC China, 2025
- Invited talk, “Memory System Design based on Coherence Interconnects”, Compunet, 2025
- Invited talk, “Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning Aggregation”, CNCC, 2024
- Invited talk, “From Emerging Applications to New Storage Technology: A Full-Stack Exploration”, Young PI forum, 2024
- Invited talk, “Progress in Near-Data Processing”, CCF Computility, 2024
- Invited talk, “Progress in Near-Data Processing”, ASSIST, 2024
- Invited talk, “Progress in Near-Data Processing”, CCF-TCIST, 2024
- Invited talk, “Progress in Near-Data Processing”, Renmin University, 2023
- Invited talk, “Storage System: From Firmware to Architecture”, Xiamen University, 2022
- Invited talk, “Revisiting the Designs of Storage Class Memory”, CCF CHIP, 2022
- Invited talk, “ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis”, Intel Computational Storage Lab, 2020
- Presentation, “ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis”, ISCA, online, 2020
- Presentation, “DRAM-less: Hardware Acceleration of Data Processing with New Memory”, HPCA, San Diego, CA, 2020
- Presentation, “Scalable Parallel Flash Firmware for Many-core Architectures”, FAST, Santa Clara, CA, 2020
- Presentation, “FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads”, HPCA, Washington DC, 2019
- Presentation, “FlashGPU: Placing New Flash Next to GPU Cores”, DAC, Las Vegas, NV, 2019
- Presentation, “Maximizing GPU Cache Utilization with Adjustable Cache Line Management”, Jeju, South Korea, 2019
- Presentation, “FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for Ultra-Low Latency SSDs”, OSDI, Carlsbad, CA, 2018
- Presentation, “FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems”, Eurosys, Porto, Portugal, 2018
- Presentation, “CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs”, IPDPS, Vancouver, Canada, 2018
- Presentation, “An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing”, NPC, Anhui, China, 2017
- Presentation, “ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture”, Inflow, Savannah, GA, 2016
- Presentation, “Couture: Tailoring STT-MRAM for Persistent Main Memory”, Inflow, Savannah, GA, 2016
- Presentation, “CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing”, ASBD, Portland, OR, 2015
- Presentation, “NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing”, PACT, San Francisco, CA, 2015
- Presentation, “Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing”, PACT SRC, San Francisco, CA, 2015
- Presentation, “OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory Characterization”, ICCD, New York city, NY, 2015
- Presentation, “Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing”, FAST WiP, Santa Clara, CA, 2015

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

-
- Presentation, "Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration", HotStorage, Philadelphia, PA, 2014
-

教学经历

- Advanced Computer Architecture (Fall'26, Fall'25, Fall'24, Fall'23)
 - Computer Architecture (Fall'26, Fall'25, Fall'24, Fall'23, Fall'22)
 - Introduction of Computer System (Fall'26, Fall'25, Fall'24, Fall'23, Fall'22)
 - IIT 3002 Operating Systems (Fall'15, Fall'16)
 - IIT 6036 Computer Organization and Design (Fall'15, Fall'16)
 - IIT 7024 Advanced System Architecture (Spring'16)
-

获奖情况

- 2026: 北京市自然科学基金青年重点项目（原北京杰青）
- 2025: HPCA 名人堂
- 2025: 奥林帕斯先锋奖
- 2023: 英特尔学术英才计划荣誉学者
- 2023: ACM SIGCSE 新星奖
- 2022: 国家自然科学基金优秀青年基金海外项目（海外优青）
- 2021: Our storage-class memory research is selected as KAIST breakthrough 50 years
- 2020-2021: Korean BK21+ Scholarship
- 2020: HPCA travel grant
- 2019: Annual Non-Volatile Memories Workshop (NVMW) -- Nominated as Memorable Paper Award
- 2019: Korea Computer Congress (KCC) -- Best Presentation Paper Award
- 2018: OSDI travel grant
- 2015: ACM Student Research Competition 2nd Runner Award